

Course Project Proposal

Knowledge Discovery and Data Mining

Project Topic: Stress-Testing Simple Graph Features under Controlled Link-Anomaly Injection

Project Type: Research

Group No.: 15

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Declaration

We hereby declare that this project is done solely within the scope of this course. The work presented in this proposal and the final project is independent and does not overlap with other concurrent courses, our Final Year Projects (FYP), or any external research projects.

Project Description

Project Overview

Randomly injected links are widely used to evaluate anomaly detection in dynamic graphs. However, they may introduce artifacts that reduce task difficulty relative to realistic anomaly detection settings. We propose a controlled empirical study of this potential evaluation bias. Using the lightweight TGF detector [1] as a fixed reference, we will vary the anomaly-generation mechanism while holding the dataset, feature extractor, classifier family, and evaluation protocol as constant as possible. The core experiment will compare random link injection with an activity-matched alternative that preserves the recent activity profile of the original endpoints. We will quantify the resulting performance change and identify the feature groups responsible for it. The intended contribution is a reproducible boundary test of the conditions under which performance based on simple graph features remains valid. The study focuses on evaluating the existing detector rather than introducing a new anomaly taxonomy or a more complex model.

Problem Statement and Research Gap

Latapy and Rajeh [1] report that trivial features computed from recent history graphs, combined with classical learning, achieve high detection performance on randomly injected links. This result has practical significance

because the detector is computationally efficient, scalable, and interpretable. Its interpretation remains conditional on the anomaly model. A uniformly sampled link may be readily distinguishable because its endpoints exhibit implausible activity levels, no prior relationship, or incompatible local neighborhoods. Consequently, a classifier may exploit artifacts of the injection procedure rather than learn a transferable representation of abnormal interaction.

Recent studies indicate that detection outcomes depend on the anomaly definition. Poštuvan et al. [2] distinguish structural, temporal, and contextual link anomalies in continuous-time dynamic graphs. AnomalyLLM [3] evaluates systematic anomaly types on UCI Messages. BAG [4] and DyGADBench [5] report substantial performance variation across synthetic, organic, localized, and temporally persistent anomalies. These works establish comprehensive taxonomies, benchmarks, or learning methods. They do not isolate, within a fixed lightweight pipeline, the change in measured performance that results from controlling a single interpretable property of the generator.

Accordingly, a focused research gap remains. Existing work lacks a paired evaluation that holds the detector and data constant, replaces random endpoint sampling with progressively more realistic constraints, and attributes the resulting performance difference to specific simple features. Addressing this gap would clarify whether TGF-style results arise from robust graph signals, generator-specific artifacts, or a combination of both.

Research Objective, Questions, and Hypotheses

The objective is to quantify the sensitivity of a simple graph-feature detector to anomaly generation and to explain that sensitivity at the feature-group level. The study addresses the following research questions:

1. **RQ1:** How much does detection performance change when random link injection is replaced by activity-matched injection?
2. **RQ2:** Which feature families account for the observed difference: endpoint activity, pair history, or local neighborhood structure?
3. **RQ3:** Is any performance loss primarily a train-test distribution shift, or are activity-matched anomalies intrinsically less separable with the available features?

We will test three corresponding hypotheses. **H1:** Activity matching lowers performance by removing an endpoint-frequency artifact. **H2:** Activity-related features lose more discriminative power than pair-history or neighborhood features. **H3:** Training on activity-matched anomalies recovers part of the performance lost when a detector trained on random anomalies is transferred to the matched setting. Complete recovery is not assumed. These hypotheses are falsifiable. A small or absent performance gap would indicate that other simple graph signals remain robust.

Proposed Methodology

Data and Reference Detector

The core study will use the public UCI Messages link stream provided as an example in the released TGF repository [1]. Each event is represented as $e_i = (t_i, u_i, v_i)$, where an interaction between nodes u_i and v_i occurs at time t_i . We will first reproduce an end-to-end TGF run and record the code version, preprocessing choices, history configuration, and random seeds.

The principal detector will use TGF history-graph features with a random forest classifier. A logistic-regression or interpretable rule-based baseline may be added to assess whether the primary result is classifier-specific. A comprehensive model comparison is outside the project scope. If part of the released pipeline is incompatible with our environment, we will retain the research question using a documented subset of strictly historical features: recent endpoint activity, prior pair frequency, and neighborhood overlap.

Controlled Anomaly Generation

We will split the benign stream chronologically into training, validation, and test periods before constructing anomalies. Any statistic used to generate or score an event at time t will depend only on events observed before t .

The reference condition will follow TGF-style random injection and retain the basic eligibility constraints of the released generator. The primary treatment will be activity-matched injection. For a selected timestamp and reference interaction, each replacement endpoint will be sampled from nodes with a similar recent activity profile. Activity may be represented by recent degree or weighted interaction count and matched through coarse quantile bins. The exact activity statistic, history length, and binning resolution will be selected using training and validation data and then fixed for final testing. This design provides methodological specificity without prematurely selecting a configuration with insufficient candidate coverage.

Where feasible, random and matched conditions will share timestamps, anomaly proportions, and seed schedules. We will examine timestamp, endpoint activity, prior pair contact, first-time versus repeated interaction status, and neighborhood distributions to identify unintended generation artifacts. An optional third condition will additionally preserve coarse community membership or local proximity. It will be included only after completion of the core activity-matched condition and confirmation of adequate candidate coverage.

Experimental Protocol

The main comparison will contain three settings:

1. Train and test on random anomalies.
2. Train on random anomalies and test on activity-matched anomalies.
3. Train and test on activity-matched anomalies.

The first setting reproduces the less constrained reference task. The second quantifies cross-generator robustness. The third distinguishes distribution-shift effects from limitations of the feature set. A fuller cross-generator matrix may be reported if it requires minimal additional computation. It is not required for the primary conclusion. All feature selection, model settings, and thresholds will be determined using training or validation data rather than the final test set.

Evaluation Plan

The primary metric will be area under the precision-recall curve (PR-AUC), since anomalous links form a minority class. We will also report ROC-AUC and recall at a pre-specified low false-positive rate. Each result will state the anomaly proportion, data split, generator, and number of seeds. Paired conditions will be summarized across repeated seeds, with uncertainty intervals where practical.

To address RQ2, we will compare feature groups or perform targeted ablations for endpoint activity, pair history, and neighborhood structure. If a compact feature subset is studied, it will use a fixed feature budget and validation-only selection. Generator diagnostics will be reported alongside predictive metrics. High predictive performance cannot be interpreted as evidence of robustness when the injection process introduces a readily detectable artifact.

Expected Contributions, Scope, and Risk Management

The project is expected to produce three contributions:

1. A controlled estimate of how strongly TGF-style performance depends on random endpoint sampling.
2. A reproducible activity-matched generator and diagnostics for detecting injection artifacts.

3. A feature-level explanation of which simple graph signals remain effective as anomalies become less random.

The minimum complete study uses one dataset, two generators, the TGF feature pipeline, and the three evaluation settings above. Community matching, MovieLens, and fixed-budget feature selection are optional extensions. This tiered scope preserves room to adapt the project after the baseline is reproduced without weakening the central research question.

The main implementation risk is insufficient candidate coverage under strict matching. If this occurs, we will coarsen activity bins or adjust the history window using validation data and report the resulting coverage. A second risk is incompatibility in the released pipeline. The fallback feature subset described above preserves the experimental logic. A null result would remain informative because it would provide evidence that the selected simple features retain discriminative value after controlling a major random-injection artifact.

Related Work

TGF constitutes the primary methodological basis of this project. Latapy and Rajeh [1] demonstrate that computationally efficient features derived from recent history graphs, combined with classical classifiers, can achieve high detection performance on randomly injected links. Their findings establish an interpretable baseline. However, the interpretation of this performance depends on the anomaly-generation process. Our project therefore subjects TGF to a targeted robustness assessment. Rather than proposing a competing detector, we examine whether its reported effectiveness persists after controlling for an endpoint-activity confound.

The closest conceptual work examines how different definitions of anomalous links affect detection. Poštuvan et al. [2] distinguish structural, temporal, and contextual anomalies in continuous-time dynamic graphs. AnomalyLLM [3] evaluates systematic anomaly types on the same UCI Messages domain used in our study. These studies provide evidence that detection performance depends on anomaly semantics. However, their primary emphasis is on comprehensive anomaly categories or learning methods. In contrast, our study holds the lightweight detector and dataset fixed and modifies one interpretable property of the generator.

Recent benchmarks further indicate that empirical conclusions vary across anomaly sources. BAG [4] compares an extensive set of models across synthetic and organic anomalies. DyGADBench [5] organizes anomalies by structural locality and temporal persistence. These benchmarks offer broad empirical coverage. However, they do not isolate the mechanisms underlying performance variation after a paired change in endpoint sampling. This distinction defines our contribution. We provide a controlled extension of the TGF study that attributes performance changes to specific feature families rather than introducing a new taxonomy, benchmark, or model.

References

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